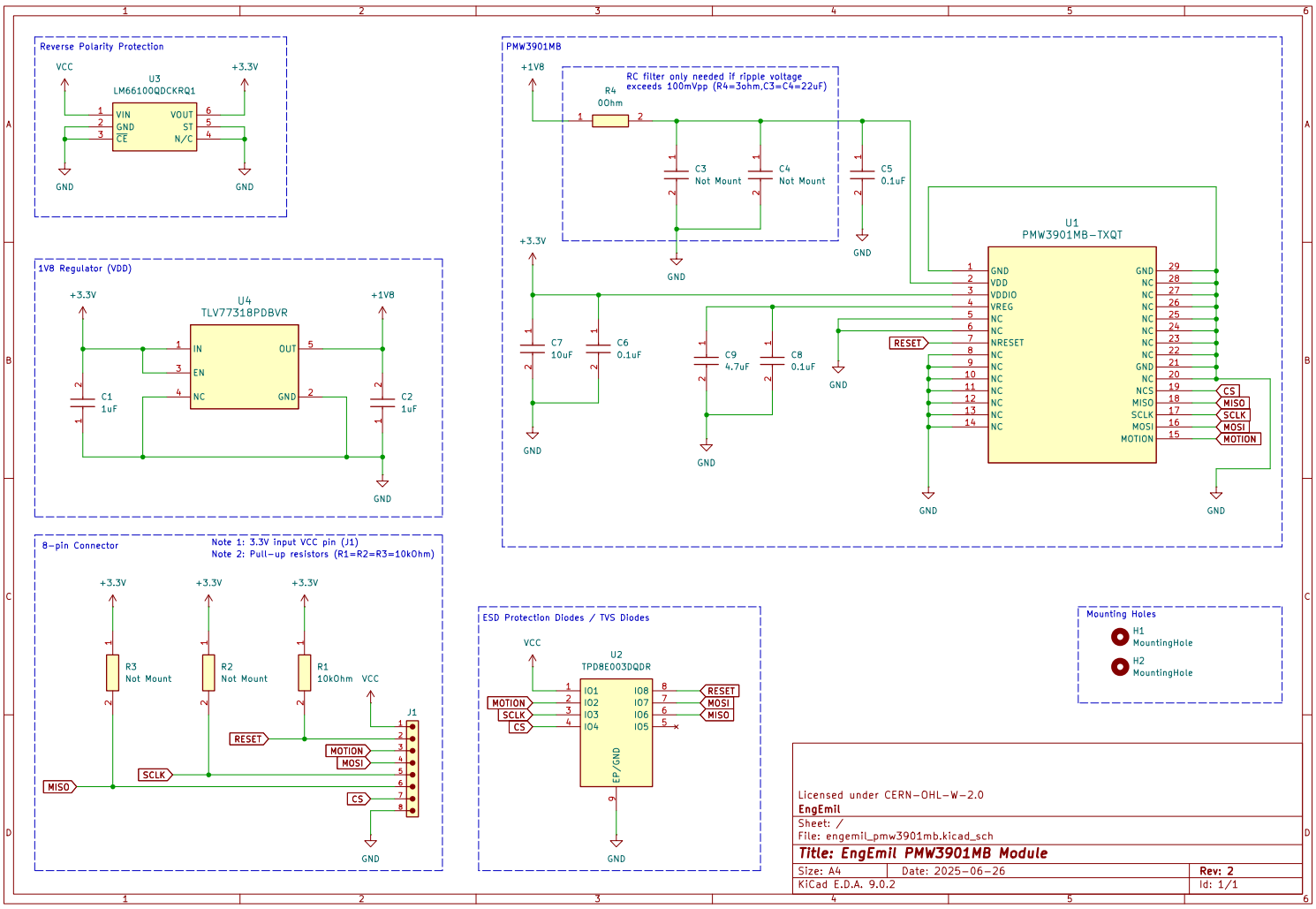
Reverse Polarity Protection
VCC
+3.3V
U3
LM66100QDCKRQ1
VIN
GND
CE
VOUT
ST
N/C
GND
GND
1V8 Regulator (VDD)
+3.3V
U4
TLV77318PDBVR
IN
EN
NC
OUT
GND
+1V8
C1
1uF
C2
1uF
GND
8-pin Connector
Note 1: 3.3V input VCC pin (J1)
Note 2: Pull-up resistors (R1=R2=R3=10kOhm)
+3.3V
+3.3V
+3.3V
R3
Not Mount
R2
Not Mount
R1
10kOhm
VCC
J1
RESET
MOTION
MOSI
SCLK
MISO
CS
GND
PMW3901MB
+1V8
RC filter only needed if ripple voltage
exceeds 100mVpp (R4=3ohm,C3=C4=22uF)
R4
0Ohm
C3
Not Mount
C4
Not Mount
C5
0.1uF
GND
+3.3V
U1
PMW3901MB-TXQT
GND
VDD
VDDIO
VREG
NC
NRESET
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NC
NCS
MISO
SCLK
MOSI
MOTION
RESET
C7
10uF
C6
0.1uF
C9
4.7uF
C8
0.1uF
GND
GND
GND
GND
CS
MISO
SCLK
MOSI
MOTION
ESD Protection Diodes / TVS Diodes
VCC
U2
TPD8E003DQDR
IO1
IO2
IO3
IO4
IO8
IO7
IO6
IO5
EP/GND
MOTION
SCLK
CS
RESET
MOSI
MISO
GND
Mounting Holes
H1
MountingHole
H2
MountingHole
Licensed under CERN-OHL-W-2.0
EngEmil
Sheet: /
File: engemil_pmw3901mb.kicad_sch
Title: EngEmil PMW3901MB Module
Size: A4
Date: 2025-06-26
Rev: 2
KiCad E.D.A. 9.0.2
Id: 1/1